

ID number: 2520090058

Task 2: To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

CODE:

```
1 144 FICD 5.09 File: km.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main() {
    char input[200];
    char name[50] = "Nikhil";
    char command[200];

    printf("Enter a command: ");
    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';

    printf("\n--- Double Quote Testing ---\n");

    /* Preserve spaces inside double quotes */
    if (strstr(input, "\"") != NULL) {
        printf("Quoted input detected.\n");
    }

    /* Variable expansion demonstration */
    printf("Variable value: %s\n", name);

    /* Parse quoted command */
    if (strcmp(input, "Hello World") == 0) {
        printf("Output: Hello World\n");
    }
    else if (strcmp(input, "My name is %s" == 0) {
        printf("Output: My name is %s\n", name);
    }
    else {
        printf("Input received: %s\n", input);
    }

    /* Test quoted command */
    printf("Command: %s\n", command);
    printf("Executing: %s\n", command);

    system(command);

    return 0;
}
```

OUTPUT:

```
Last login: Thu Aug 28 14:55:02 on ttys000
greenikhilreddy@Strees-MacBook-Pro ~ % nano km.c
greenikhilreddy@Strees-MacBook-Pro ~ % gcc km.c -o km
clang: error: unknown argument: '-o'
clang: error: no such file or directory: 'km'
greenikhilreddy@Strees-MacBook-Pro ~ % gcc km.c -o km
greenikhilreddy@Strees-MacBook-Pro ~ % ./km
Enter a command: ls
--- Double Quote Testing ---
Variable value: Nikhil
Input received: ls
Executing: echo "Hello World"
Hello World
greenikhilreddy@Strees-MacBook-Pro ~ %
```

SESSION-6

Task 1: To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

CODE:

```
LM FICD 3.09 File: ad.c
#include <stdio.h>
#include <string.h>
#define MAX 200

void processEscape(char input[]) {
    char output[MAX];
    int i, j = 0;

    for (i = 0; input[i] != '\0'; i++) {
        if (input[i] == '\\') {
            i++;
            if (input[i] != '\0') {
                output[j++] = input[i];
            }
        } else {
            output[j++] = input[i];
        }
    }

    output[j] = '\0';
    printf("Parsed Output: %s\n", output);
}

int main() {
    char input[MAX];

    printf("Enter input: ");
    fgets(input, MAX, stdin);
    input[strlen(input) - 1] = '\0';
    printf("Original Input: %s\n", input);
    processEscape(input);
    return 0;
}
```

OUTPUT:

```
Last login: Thu Aug 20 14:59:00 on tty000
sreeni@hilleddy@Stress-MacBook-Pro ~ % nano ad.c
sreeni@hilleddy@Stress-MacBook-Pro ~ % gcc nano.c -o nano
clang: error: no such file or directory: 'nano.c'
clang: error: no input files
sreeni@hilleddy@Stress-MacBook-Pro ~ % gcc ad.c -o ad
sreeni@hilleddy@Stress-MacBook-Pro ~ % ./ad
Enter input: ls

Original Input: ls
Parsed Output: ls
sreeni@hilleddy@Stress-MacBook-Pro ~ % ./ad
Enter input: ls

Original Input: ls
Parsed Output: ls
sreeni@hilleddy@Stress-MacBook-Pro ~ %
```

Task 2: To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching

CODE:

```
1  LW PIC0 3.09
2
3  #include <stdio.h>
4  #include <stdlib.h>
5  #include <unistd.h>
6  #include <sys/wait.h>
7
8  int main() {
9      pid_t pid;
10     int status;
11
12     printf("Parent Process Started\n");
13     printf("Parent PID: %d\n", getpid());
14
15     /* Create child process */
16     pid = fork();
17
18     if (pid < 0) {
19         perror("fork failed");
20         return 1;
21     }
22
23     if (pid == 0) {
24         /* Child process */
25         printf("\nChild Process Created\n");
26         printf("Child PID: %d\n", getpid());
27         printf("Parent PID: %d\n", getppid());
28
29         /* Execute program with arguments */
30         printf("Executing: ls -l\n");
31
32         execvp("ls", "ls", "-l", NULL);
33
34         /* Executes only if execvp fails */
35         perror("Execution failed");
36         exit(1);
37     }
38     else {
39         /* Parent process */
40         printf("Parent waiting for child...\n");
41
42         if (waitpid(pid, &status, 0) == -1) {
43             perror("waitpid failed");
44             return 1;
45         }
46
47         if (WEXITED(status)) {
48             printf("\nChild exited with status: %d\n",
49                    WEXITSTATUS(status));
50         }
51         else {
52             printf("\nChild terminated abnormally\n");
53         }
54
55         printf("Parent Process Finished\n");
56     }
57
58     return 0;
59 }
```

OUTPUT:

```
Parent waiting for child...
Child Process Created
Child PID: 1247
Parent PID: 1246
Executing: ls -l
total 1648
-rw-r--r-- 1 sreenishreddy staff 286 15 Aug 11:31 88.c
-rw-r--r-- 1 sreenishreddy staff 183 25 Jul 09:24 1.c.save
-rwx-r-x 1 sreenishreddy staff 9888 15 Aug 11:00 180
-rw-r--r-- 1 sreenishreddy staff 1189 15 Aug 11:02 180.c
-rwx-r-x 1 sreenishreddy staff 8776 15 Aug 11:05 181
-rw-r--r-- 1 sreenishreddy staff 1241 15 Aug 11:05 181.c
-rwx-r-x 1 sreenishreddy staff 8776 15 Aug 11:07 182
-rw-r--r-- 1 sreenishreddy staff 1241 15 Aug 11:07 111.c
-rwx-r-x 1 sreenishreddy staff 8736 15 Aug 12:03 1111
-rw-r--r-- 1 sreenishreddy staff 1819 15 Aug 12:03 1111.c
-rwx-r-x 1 sreenishreddy staff 8776 15 Aug 13:03 11111
-rw-r--r-- 1 sreenishreddy staff 1883 15 Aug 13:28 11111.c
-rwx-r-x 1 sreenishreddy staff 8408 25 Jul 09:39 12
-rw-r--r-- 1 sreenishreddy staff 183 25 Jul 09:40 12.c
-rwx-r-x 1 sreenishreddy staff 8736 15 Aug 12:04 123
-rw-r--r-- 1 sreenishreddy staff 1849 15 Aug 13:45 123.c
-rwx-r-x 1 sreenishreddy staff 1849 15 Aug 18:59 123.c.save
-rw-r--r-- 1 sreenishreddy staff 8028 15 Aug 12:11 1234
-rwx-r-x 1 sreenishreddy staff 307 11 Aug 12:11 1234.c
-rwx-r-x 1 sreenishreddy staff 8736 15 Aug 12:23 12344
-rw-r--r-- 1 sreenishreddy staff 888 15 Aug 12:31 12344.c
-rw-r--r-- 1 sreenishreddy staff 186 15 Aug 12:23 12344.c.save
-rwx-r-x 1 sreenishreddy staff 2746 15 Aug 12:23 1234567.c
-rw-r--r-- 1 sreenishreddy staff 1842 15 Aug 12:27 16.c
-rwx-r-x 1 sreenishreddy staff 8028 1 Aug 08:28 2
-rwx-r-x 1 sreenishreddy staff 816 1 Aug 08:15 2.c
-rwx-r-x 1 sreenishreddy staff 8696 28 Aug 12:14 22
-rw-r--r-- 1 sreenishreddy staff 1318 28 Aug 12:14 22.c
-rwx-r-x 1 sreenishreddy staff 8784 15 Aug 12:26 222
-rw-r--r-- 1 sreenishreddy staff 918 15 Aug 12:26 222.c
-rwx-r-x 1 sreenishreddy staff 8784 15 Aug 12:00 2222
-rw-r--r-- 1 sreenishreddy staff 389 18 Aug 12:00 2222.c
-rwx-r-x 1 sreenishreddy staff 8936 28 Aug 12:15 23
-rw-r--r-- 1 sreenishreddy staff 1291 28 Aug 12:16 23.c
-rwx-r-x 1 sreenishreddy staff 8848 28 Aug 12:21 24
-rw-r--r-- 1 sreenishreddy staff 1471 28 Aug 12:21 24.c
-rwx-r-x 1 sreenishreddy staff 8648 18 Aug 12:14 243
-rw-r--r-- 1 sreenishreddy staff 86 15 Aug 12:14 243.c
-rwx-r-x 1 sreenishreddy staff 13528 28 Aug 12:25 25
-rw-r--r-- 1 sreenishreddy staff 2627 28 Aug 12:25 25.c
-rwx-r-x 1 sreenishreddy staff 8776 15 Aug 11:47 264
-rw-r--r-- 1 sreenishreddy staff 1458 15 Aug 11:47 264.c
-rwx-r-x 1 sreenishreddy staff 9888 1 Aug 08:56 3
-rw-r--r-- 1 sreenishreddy staff 1136 1 Aug 08:27 3.c
-rwx-r-x 1 sreenishreddy staff 688 28 Aug 18:56 31.c
-rwx-r-x 1 sreenishreddy staff 8792 18 Aug 12:28 333
-rw-r--r-- 1 sreenishreddy staff 813 18 Aug 12:27 333.c
-rwx-r-x 1 sreenishreddy staff 8784 15 Aug 12:29 334
-rw-r--r-- 1 sreenishreddy staff 787 18 Aug 12:43 334.c
-rwx-r-x 1 sreenishreddy staff 8736 1 Aug 09:41 34
-rw-r--r-- 1 sreenishreddy staff 478 8 Aug 09:41 34.c
-rw-r--r-- 1 sreenishreddy staff 1 15 Aug 11:08 35.c.save
-rwx-r-x 1 sreenishreddy staff 8928 15 Aug 13:28 398
-rw-r--r-- 1 sreenishreddy staff 1735 15 Aug 13:28 398.c
-rwx-r-x 1 sreenishreddy staff 8744 1 Aug 08:50 4
-rw-r--r-- 1 sreenishreddy staff 588 4 Aug 12:31 4.c
-rwx-r-x 1 sreenishreddy staff 626 4 Aug 12:25 4.c.save
-rw-r--r-- 1 sreenishreddy staff 1148 15 Aug 18:57 45.c
-rwx-r-x 1 sreenishreddy staff 8880 15 Aug 12:23 456
-rw-r--r-- 1 sreenishreddy staff 738 15 Aug 12:23 456.c
-rwx-r-x 1 sreenishreddy staff 8784 28 Aug 14:51 467
-rw-r--r-- 1 sreenishreddy staff 1848 28 Aug 14:51 467.c
-rwx-r-x 1 sreenishreddy staff 8648 15 Aug 11:59 567
```

```
-W-r----- 1 screenhiredy staff 688 28 Aug 15:04 sd.c
dwxr-xr-x 7 screenhiredy staff 221 11 Aug 11:58 Applications
dwxr-xr-x 4 screenhiredy staff 128 31 Jul 09:13 Applications (Parallel)
-W-r----- 1 screenhiredy staff 874 28 Aug 15:04
-W-r----- 1 screenhiredy staff 1248 28 Aug 15:04 c.c
-W-r----- 1 screenhiredy staff 8776 18 Aug 12:04 cc
-W-r----- 1 screenhiredy staff 444 18 Aug 12:03 cc.c
-W-r----- 1 screenhiredy staff 1241 15 Aug 11:26 copy
dwxr-xr-x 3 root 36 14 Aug 14:52 data
dwxr-xr-x 8 screenhiredy staff 206 28 Aug 15:03 Desktop
dwxr-xr-x 2 screenhiredy staff 64 1 Aug 09:41 dir1
dwxr-xr-x 2 screenhiredy staff 64 1 Aug 09:41 dir2
dwxr-xr-x 2 screenhiredy staff 64 1 Aug 09:41 dir3
dwxr-xr-x 38 screenhiredy staff 968 28 Aug 14:45 Documents
dwxr-xr-x 126 screenhiredy staff 4832 28 Aug 14:48 Downloads
-W-r----- 1 screenhiredy staff 643 18 Aug 12:04 file.c
-W-r----- 1 screenhiredy staff 643 18 Aug 12:04 file.c.save
-W-r----- 1 screenhiredy staff 64 1 Aug 09:41 file2.txt
-W-r----- 1 screenhiredy staff 121 15 Aug 12:32 gh.c
-W-r----- 1 screenhiredy staff 1553 28 Aug 14:59 gt.c
dwxr-xr-x 13 screenhiredy staff 436 15 Aug 15:38 helloworld
-W-r----- 1 screenhiredy staff 436 18 Aug 13:23 hl.c
dwxr-xr-x 1 screenhiredy staff 8888 15 Aug 12:04 ka
-W-r----- 1 screenhiredy staff 883 15 Aug 12:04 ka.c
-W-r----- 1 screenhiredy staff 884 15 Aug 12:04 ka.c.save
-W-r----- 1 screenhiredy staff 1593 15 Aug 12:12 ka.c.save
-W-r----- 1 screenhiredy staff 1789 18 Aug 13:25 al.c.save
dwxr-xr-x 1 screenhiredy staff 8936 28 Aug 14:04 km
-W-r----- 1 screenhiredy staff 1816 28 Aug 14:58 km.c
dwxr-xr-x 97 screenhiredy staff 3166 28 Aug 14:28 library
-W-r----- 1 screenhiredy staff 2727 16 Aug 12:11 lk.c
-W-r----- 1 screenhiredy staff 1611 28 Aug 15:04 ll.c.save
dwxr-xr-x 1 screenhiredy staff 8448 15 Aug 12:16 llll
-W-r----- 1 screenhiredy staff 127 15 Aug 12:16 llll.c
-W-r----- 1 screenhiredy staff 2168 28 Aug 14:56 ml.c.save
dwxr-xr-x 7 screenhiredy staff 224 6 May 11:15 Movies
dwxr-xr-x 8 screenhiredy staff 168 17 Dec 2003 Music
dwxr-xr-x 2 screenhiredy staff 64 1 Aug 09:48 myfolder
-W-r----- 1 screenhiredy staff 1553 28 Aug 14:58 memo.c.save
dwxr-xr-x 2 screenhiredy staff 64 17 Oct 2003 NewFolderName
-W-r----- 1 screenhiredy staff 1982 15 Aug 12:14 nlk.c.save
dwxr-xr-x 3 screenhiredy staff 36 1 Aug 08:03 Parallels
dwxr-xr-x 8 screenhiredy staff 206 6 May 11:15 Pictures
dwxr-xr-x 1 screenhiredy staff 8736 15 Aug 11:38 process
-W-r----- 1 screenhiredy staff 1494 15 Aug 11:34 process.c
-W-r----- 1 screenhiredy staff 1495 15 Aug 11:37 process.c.save
dwxr-xr-x 3 screenhiredy staff 96 2 Jan 15:16 PROJECT_SDA_3
dwxr-xr-x 3 screenhiredy staff 96 1 Aug 09:41 projects
-W-r----- 1 screenhiredy staff 8746 15 Aug 12:02 pu
-W-r----- 1 screenhiredy staff 839 18 Aug 12:28 pu.c
dwxr-xr-x 4 screenhiredy staff 128 38 Aug 2003 Public
dwxr-xr-x 1 screenhiredy staff 8776 15 Aug 11:29 pwd
-W-r----- 1 screenhiredy staff 1241 15 Aug 11:29 pwd.c
-W-r----- 1 screenhiredy staff 175 15 Aug 11:29 pwd.c.save
dwxr-xr-x 9 screenhiredy staff 288 17 Mar 18:41 python
-W-r----- 1 screenhiredy staff 64 15 Aug 11:08 source.txt
dwxr-xr-x 1 screenhiredy staff 9888 7 Aug 14:48 task
-W-r----- 1 screenhiredy staff 1151 7 Aug 14:48 task.c
-W-r----- 1 screenhiredy staff 6175 4 Sep 2003 terminal
-W-r----- 1 screenhiredy staff 1432 28 Aug 14:55 tt.c.save
dwxr-xr-x 1 screenhiredy staff 8656 15 Aug 12:11 we
-W-r----- 1 screenhiredy staff 554 18 Aug 12:21 we.c
dwxr-xr-x 1 screenhiredy staff 8656 8 Aug 09:11 week3
dwxr-xr-x 1 screenhiredy staff 8832 8 Aug 08:30 week2
-W-r----- 1 screenhiredy staff 881 8 Aug 08:30 week2.c
dwxr-xr-x 1 screenhiredy staff 8458 8 Aug 09:06 week2skill
-W-r----- 1 screenhiredy staff 341 8 Aug 09:06 week2skill.c
dwxr-xr-x 1 screenhiredy staff 8656 8 Aug 08:21 week3
-W-r----- 1 screenhiredy staff 378 8 Aug 09:18 week3skill.c
-W-r----- 1 screenhiredy staff 556 18 Aug 12:05 we.c
```

```
Child exited with status: 0
Parent Process Finished
screenhiredy@rees-MacBook-Pro ~ %
```